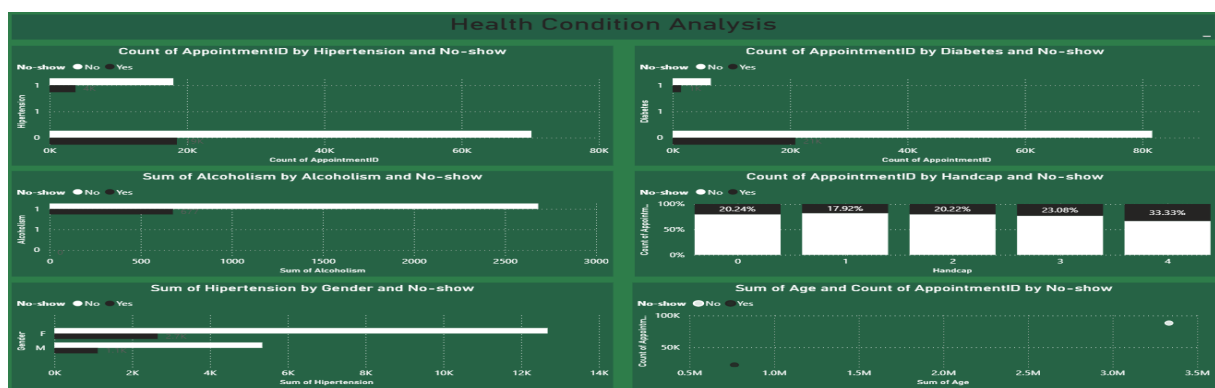


Healthcare Appointment No-Show Prediction

A Data Analytics and Machine Learning Project Report



Project Report | June 2026

1. Project Objective

Missed medical appointments (no-shows) are a persistent challenge for healthcare providers, leading to wasted clinical capacity, increased operating costs, and longer waiting lists for other patients. The objective of this project is to analyze a real-world healthcare appointment dataset to understand the factors that drive patient no-shows, build an interactive analytics dashboard for operational monitoring, and develop a machine learning classification model capable of predicting whether a patient will miss a scheduled appointment. The end goal is to translate these insights into actionable recommendations that can help clinics reduce no-show rates.

1.1 Key Goals

- Perform exploratory data analysis (EDA) to uncover patterns in patient demographics and behavior.
- Identify the strongest predictors of appointment no-shows (e.g. waiting time, age, SMS reminders).
- Build interactive Power BI dashboards for executives and operations teams.
- Train and evaluate a Decision Tree classification model to predict no-shows.
- Provide data-driven recommendations to reduce the no-show rate.

2. Dataset Summary

The dataset consists of approximately 110,527 medical appointment records from public healthcare facilities in Brazil, spanning April to June. Each record represents a single scheduled appointment along with patient demographic information, health conditions, and whether the patient attended (No-show = 0) or missed (No-show = 1) the appointment.

Field	Description
PatientId / AppointmentID	Unique identifiers for patient and appointment
Gender	Patient gender (F / M)
Age	Patient age in years
Neighbourhood	Location of the healthcare facility
Scholarship	Enrollment in Brazil's welfare program (0/1)
Hipertension, Diabetes, Alcoholism, Homocysteine	Chronic health condition flags
SMS_received	Whether an SMS reminder was sent (0/1)
Weekday / Month	Engineered date features from appointment day
WaitingDays	Engineered feature: days between scheduling and appointment
No-show	Target variable: 0 = attended, 1 = missed appointment

2.1 Class Balance

Of the 110,527 appointments, 88,208 (79.81%) were attended and 22,319 (20.19%) were no-shows, indicating a moderately imbalanced target class that was an important consideration during model evaluation.

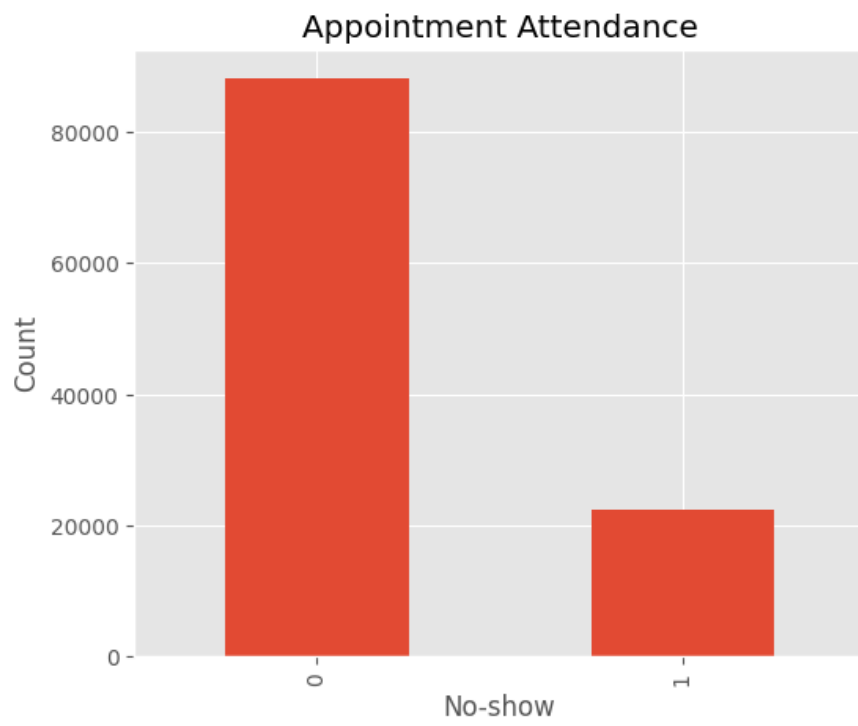


Figure 1: Distribution of appointment attendance (0 = Attended, 1 = No-show)

3. Exploratory Data Analysis (EDA)

A detailed exploratory analysis was performed to understand patient demographics, appointment scheduling behavior, and how these relate to no-show outcomes.

3.1 Demographic Patterns

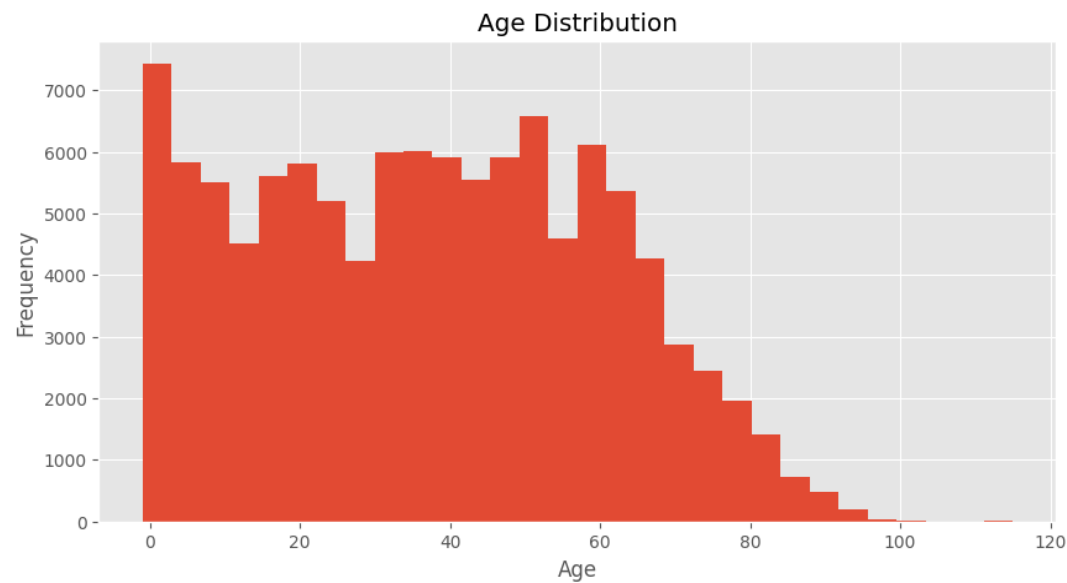


Figure 2: Age distribution of patients shows a wide spread across all age groups, with a notable spike for infants (age 0).

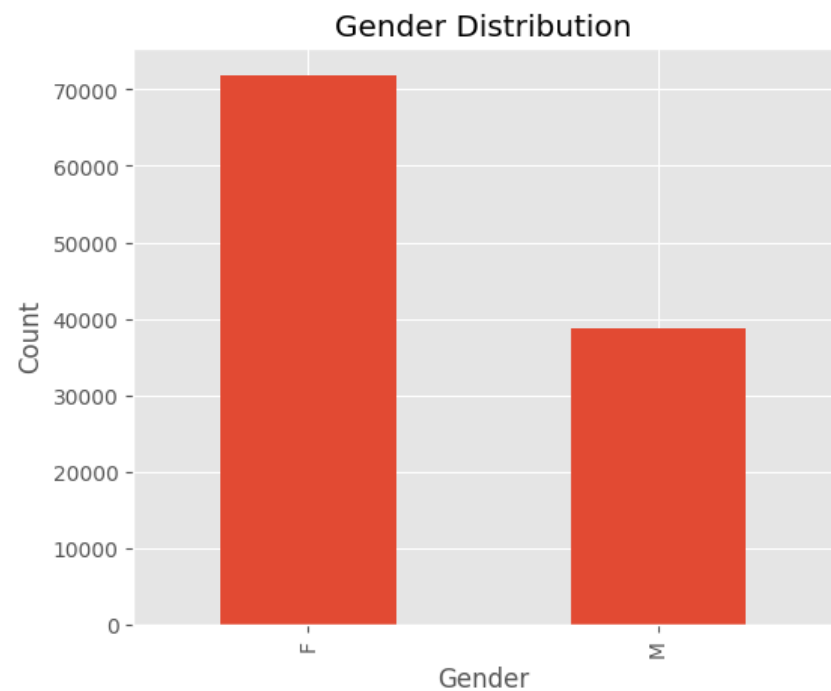


Figure 3: Female patients (about 65%) make up the majority of appointments compared to male patients.

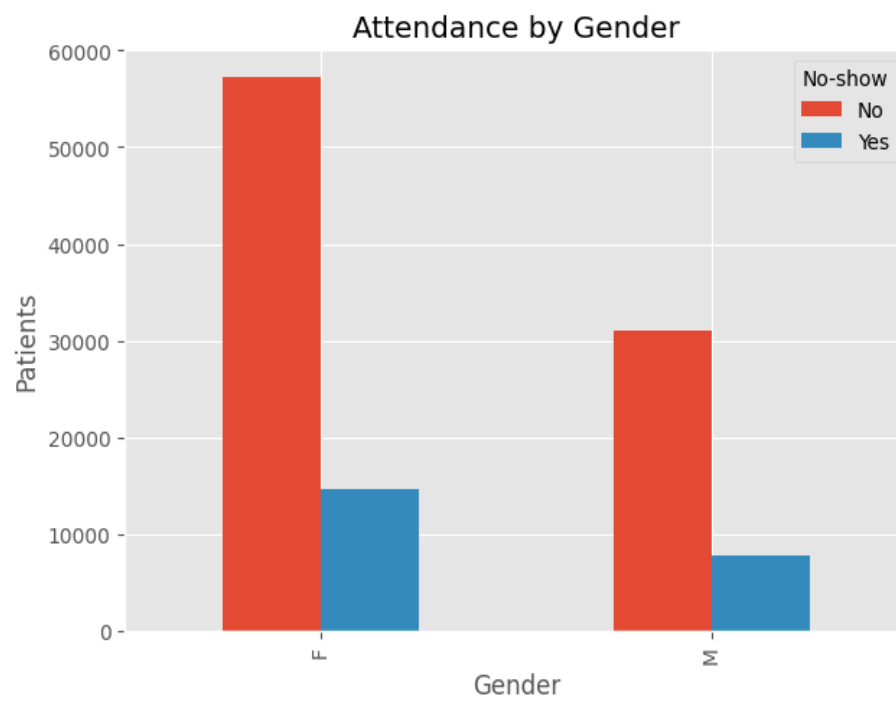


Figure 4: No-show rates are broadly similar across genders, suggesting gender is a weak standalone predictor.

3.2 Behavioral and Operational Factors

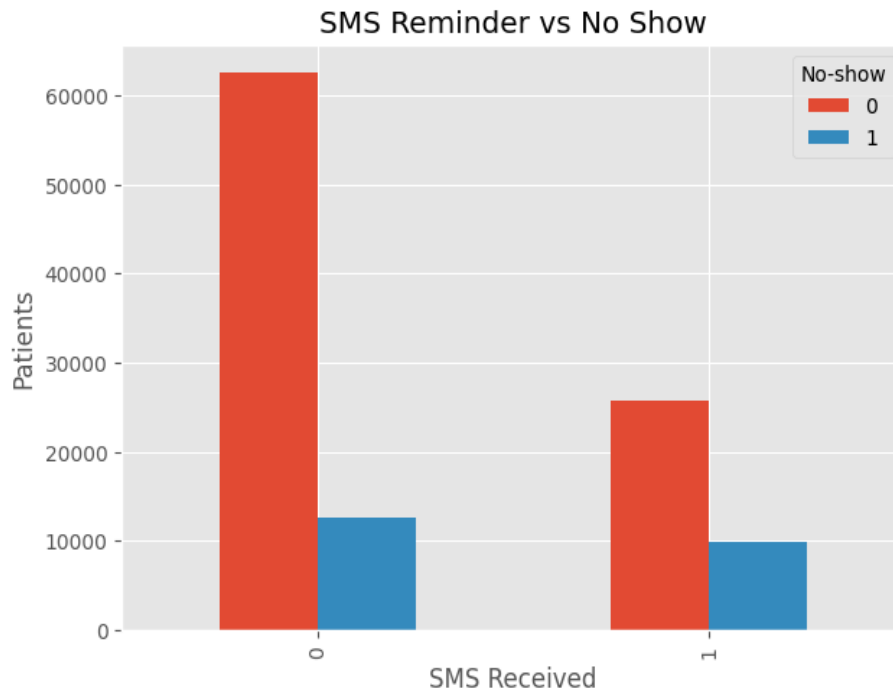


Figure 5: Counter-intuitively, patients who received an SMS reminder show a higher proportion of no-shows, likely because reminders are more often sent for appointments booked with longer lead times (which independently increase no-show risk).

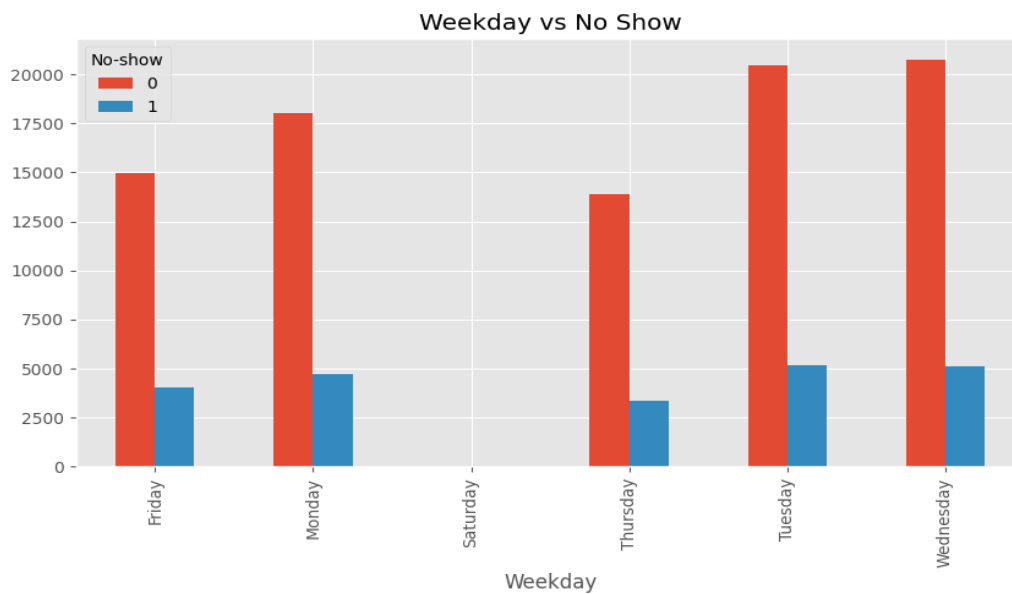


Figure 6: Appointment volumes and no-show counts are highest on Tuesday and Wednesday, with no appointments recorded on Saturday.

3.3 Health Conditions and Socioeconomic Factors

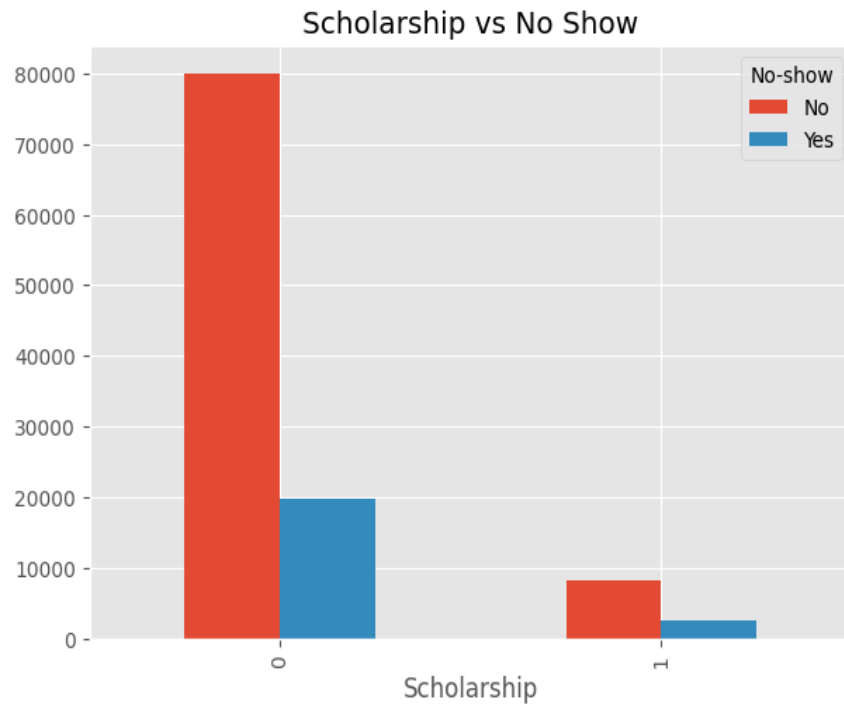


Figure 7: Patients enrolled in the welfare scholarship program show a slightly elevated no-show rate.

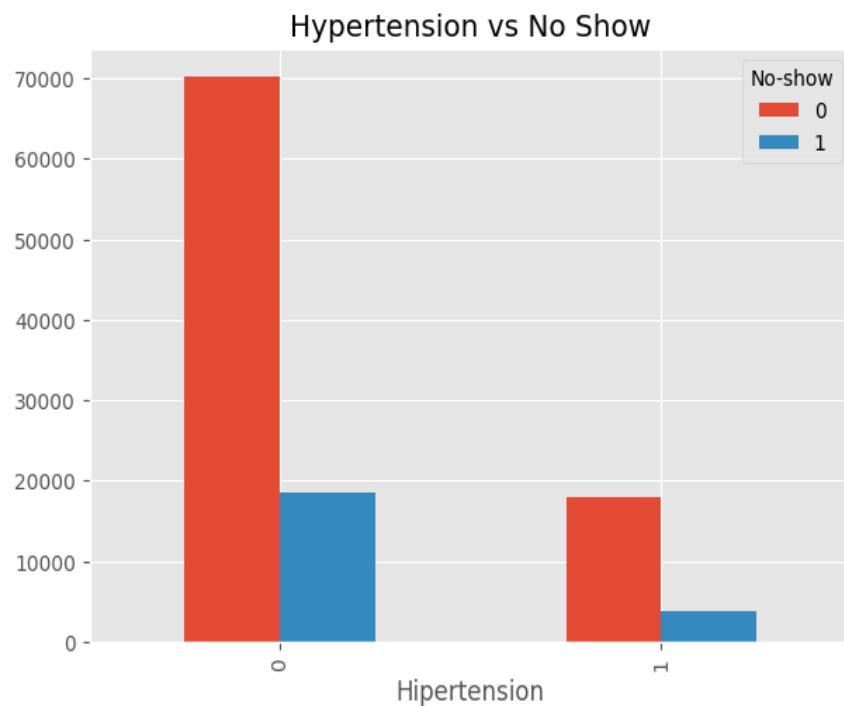


Figure 8: Patients with hypertension show a marginally lower proportion of no-shows, possibly reflecting greater engagement with ongoing care.

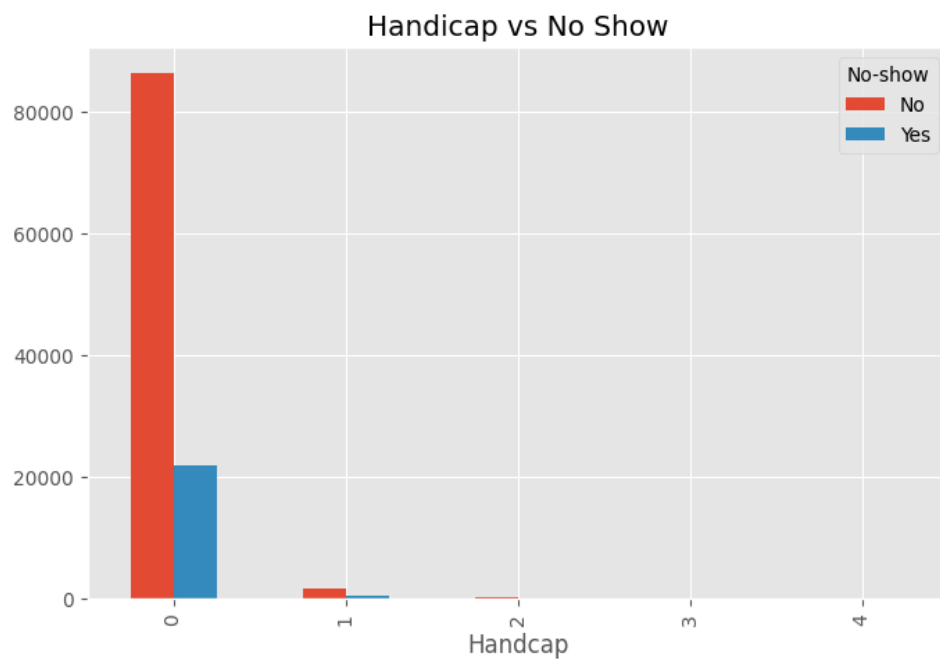


Figure 9: The vast majority of patients have no handicap; no-show proportions rise slightly with handicap severity level.

3.4 Geographic and Correlation Analysis

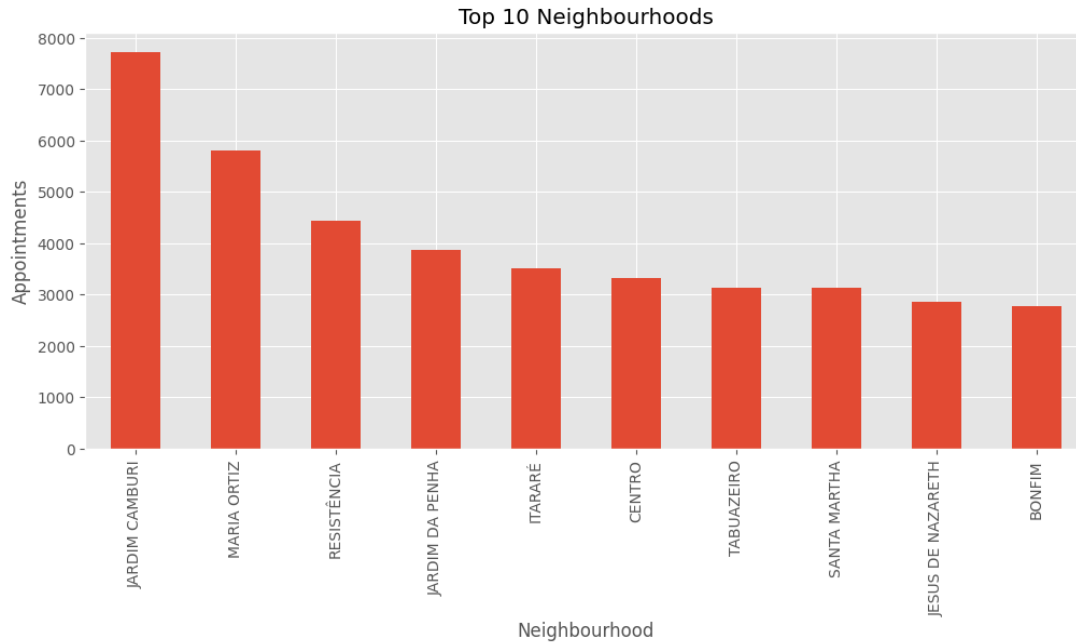


Figure 10: Jardim Camburi, Maria Ortiz, and Resistência are the top three neighbourhoods by appointment volume.

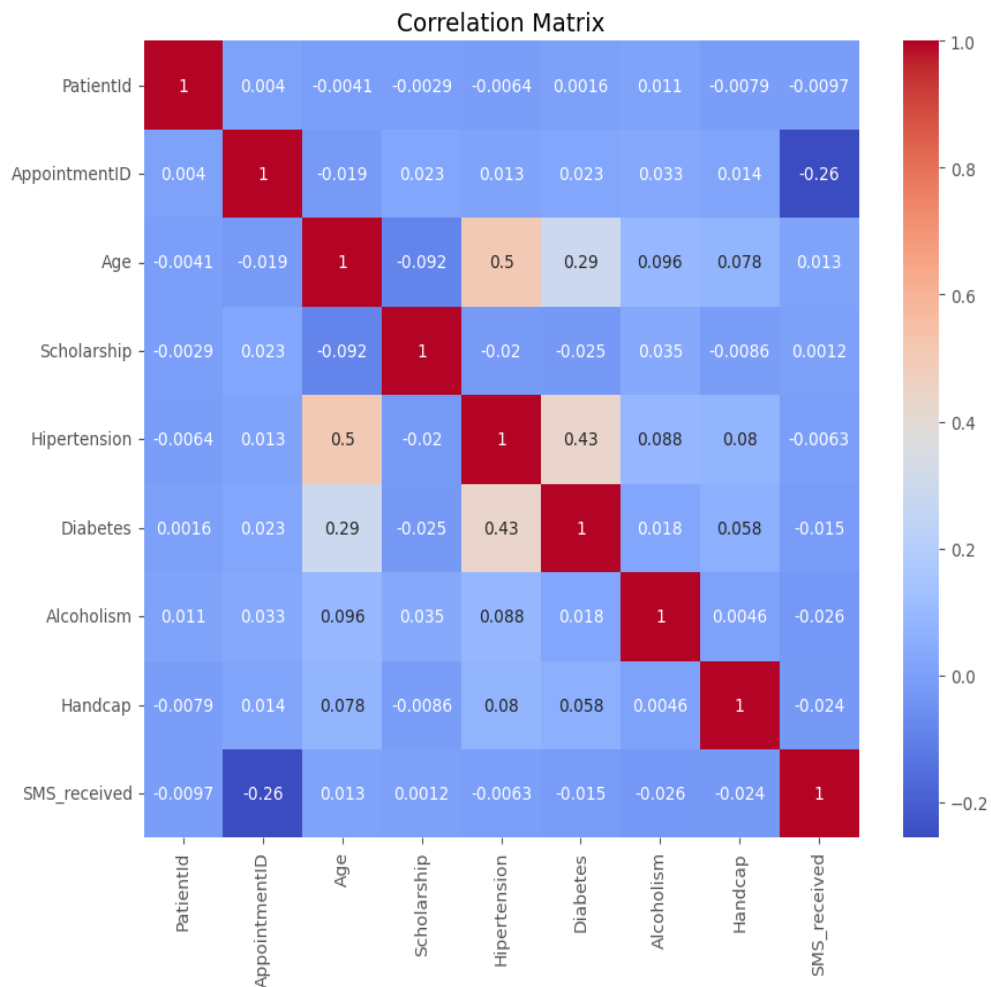


Figure 11: Correlation matrix of numerical features. Age correlates moderately with Hypertension (0.50) and Diabetes (0.29), as expected clinically. Most features show weak pairwise correlation, indicating no single variable dominates the dataset.

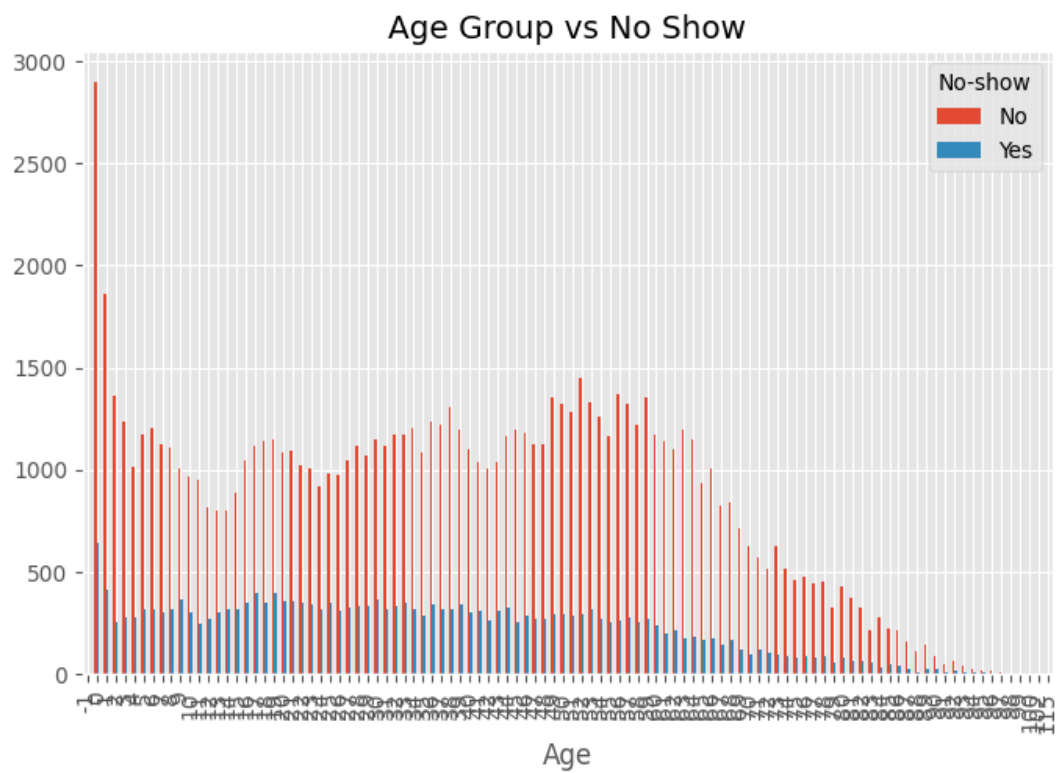


Figure 12: No-show counts by detailed age, showing relatively higher no-show proportions among younger adult age groups.

4. Machine Learning Model Performance

A Decision Tree Classifier was trained to predict patient no-shows using engineered features including WaitingDays (days between booking and appointment), Age, Gender, Neighbourhood, Weekday, SMS_received, and chronic health condition flags. The dataset was split into training and test sets, and the model was evaluated using standard classification metrics as well as 5-fold cross-validation.

4.1 Evaluation Metrics

Metric	Value
Accuracy	0.7979
Precision (No-show class)	0.3529
Recall (No-show class)	0.0013
F1 Score (No-show class)	0.0027
AUC Score	0.7199
Cross-Validation Accuracy	0.7960

The model achieves a strong overall accuracy of 79.79%, comparable to simply predicting the majority class (attended) for every patient. The classification report confirms this imbalance: the model recalls 100% of the 'Attended' class but only 0.13% of the 'No-show' class, meaning it almost never successfully flags a patient who will actually miss their appointment. The confusion matrix below illustrates this clearly, with only 9 true no-shows correctly identified out of 4,463.

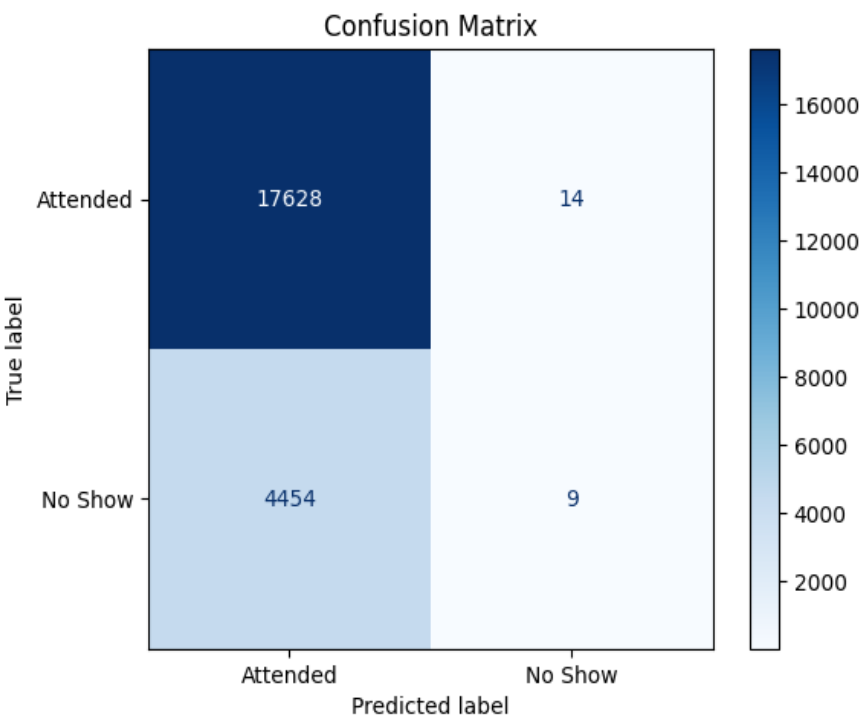


Figure 13: Confusion matrix on the test set — the model is heavily biased toward predicting 'Attended'.

4.2 Feature Importance

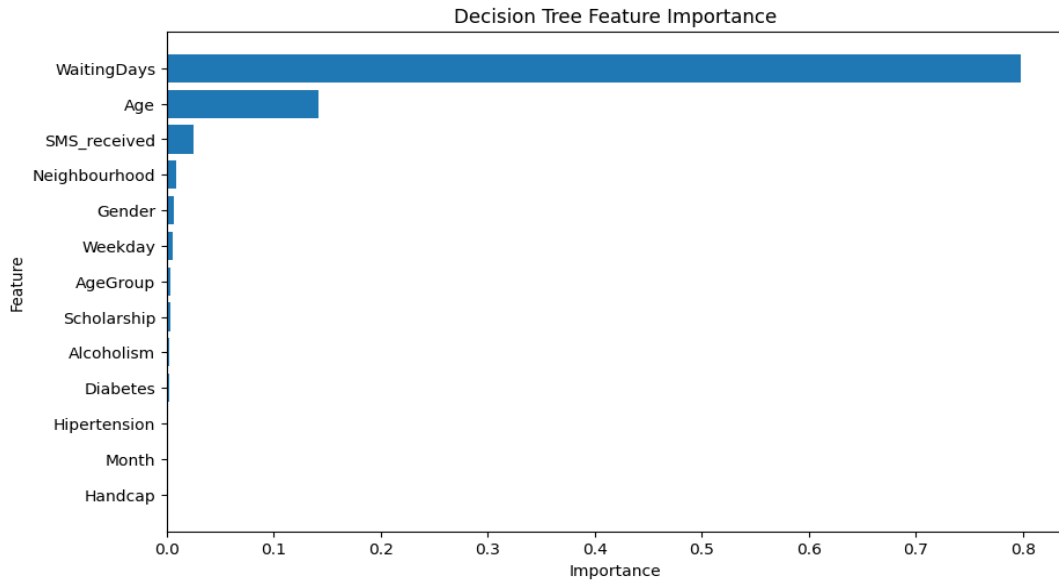


Figure 14: *WaitingDays* is overwhelmingly the most important feature (importance ≈ 0.80), followed distantly by *Age* (≈ 0.14) and *SMS_received* (≈ 0.03). Health conditions and scheduling month contribute negligibly to the model's decisions.

4.3 ROC Curve

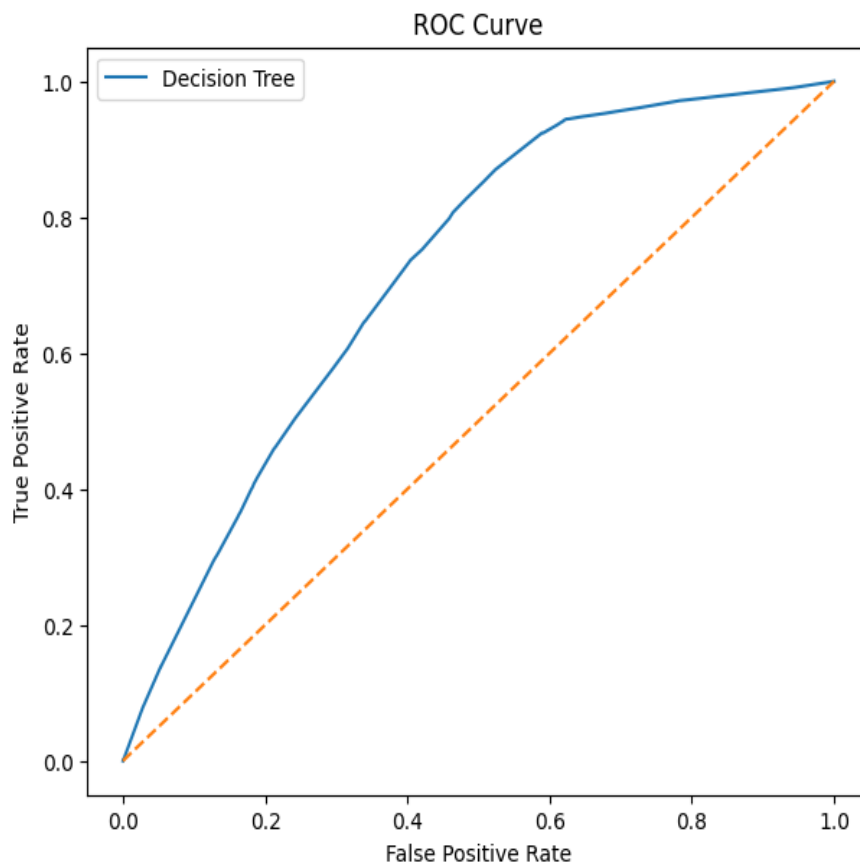


Figure 15: The ROC curve (AUC = 0.72) shows the model has moderate discriminative power between classes overall, despite the very low recall at the default 0.5 probability threshold.

4.4 Decision Tree Structure

The visualized decision tree confirms that the very first and most decisive split in the model is $\text{WaitingDays} \leq 0.5$ days, separating same-day appointments (very low no-show risk) from appointments booked in advance. Subsequent splits refine predictions using Age, Neighbourhood, and SMS_received.

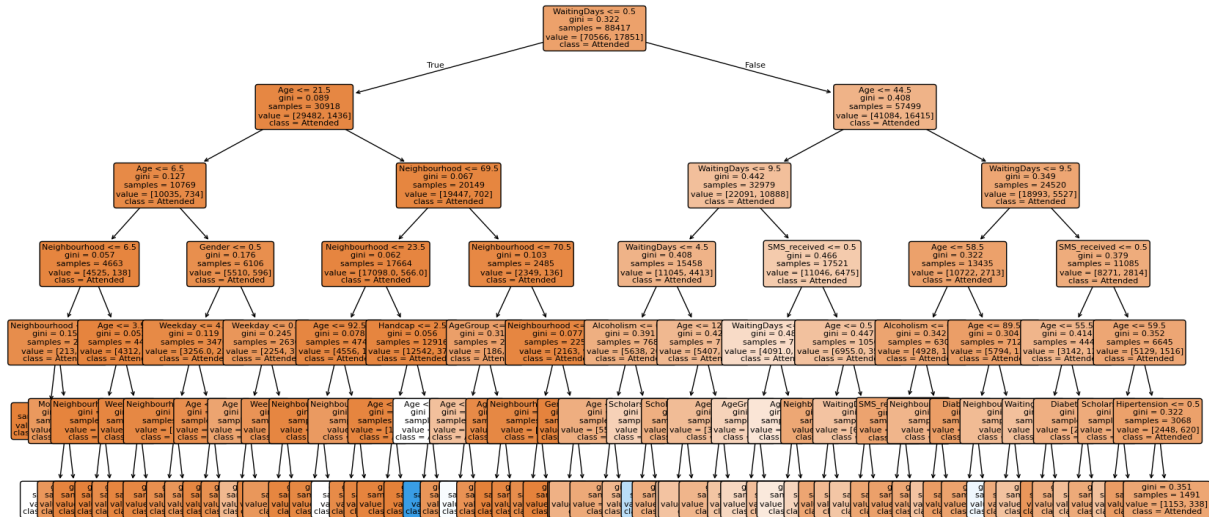


Figure 16: Visualization of the trained Decision Tree (top levels), rooted at WaitingDays.

5. Power BI Dashboards

Three interactive Power BI dashboards were developed to support different stakeholder needs: an executive summary view, an operational appointment-scheduling view, and a clinical health-condition view.

5.1 Executive Dashboard

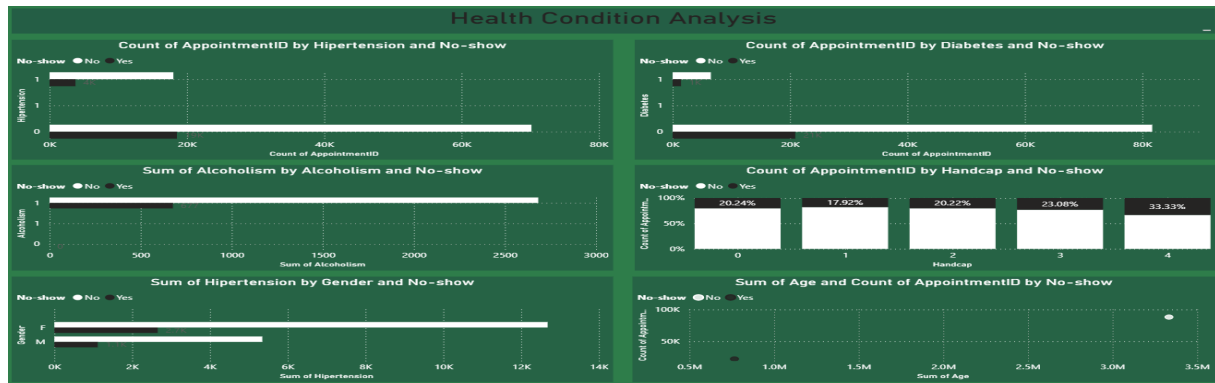


Figure 17: The Executive Dashboard summarizes total appointments (110.5K), patients attended (88.2K), no-show patients (22.3K), and an overall attendance rate, alongside breakdowns by gender and scholarship status.

5.2 Appointment Analysis Dashboard

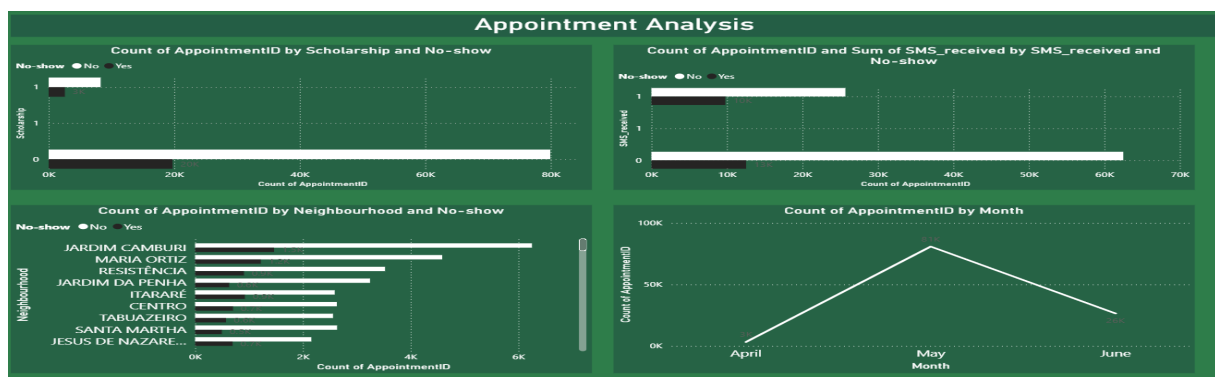


Figure 18: This dashboard tracks scholarship status, SMS reminders, top neighbourhoods, and monthly appointment volume trends, which peak in May.

5.3 Health Condition Analysis Dashboard

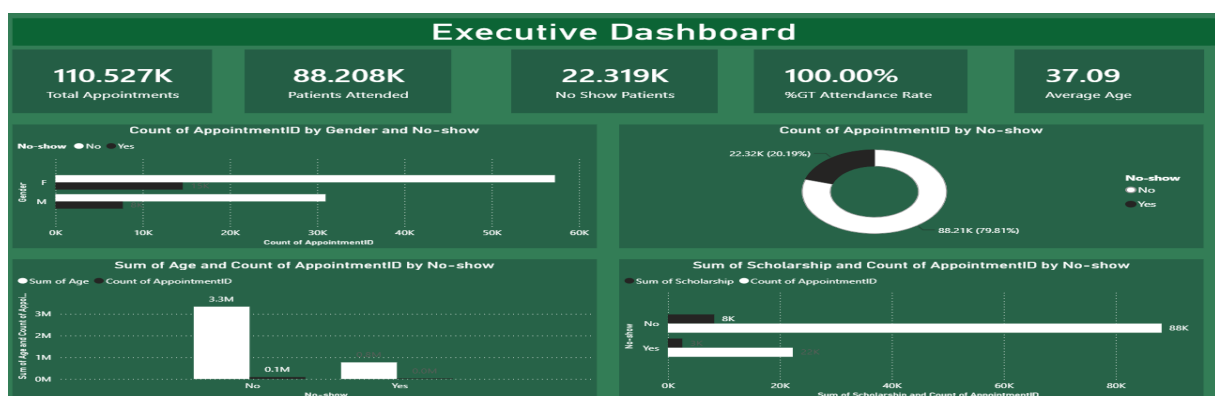


Figure 19: This dashboard breaks down no-show behavior by hypertension, diabetes, alcoholism, handicap level, gender, and age, supporting clinical resource planning.

5.4 Dashboard Insights

Beyond the static visuals, the interactive Power BI dashboards surface several operational insights that directly support day-to-day decision-making for clinic administrators, schedulers, and clinical staff.

Executive Dashboard — Insights

- Of 110,527 total scheduled appointments, 88,208 (79.81%) were attended, confirming that roughly 1 in 5 appointments is lost to no-shows, a meaningful capacity gap for resource planning.
- Female patients account for nearly double the appointment volume of male patients (≈57K vs ≈31K attended), so any clinic-wide intervention will disproportionately affect the female patient segment in absolute terms.
- The average patient age of 37.09 years indicates a working-age-dominant patient base, reinforcing the EDA finding that younger adults need targeted attention.
- The scholarship breakdown shows attendance is heavily concentrated among non-scholarship patients (88K), while scholarship patients, though fewer (22K total with scholarship+no-show), still warrant equity-focused outreach given their slightly higher no-show propensity.

Appointment Analysis Dashboard — Insights

- Appointment volume peaks sharply in May (≈81K) compared to April (≈3K) and June (≈26K), suggesting seasonal or scheduling-cycle effects that should inform staffing levels month-to-month.
- SMS reminders were sent for the majority of appointments (over 60K marked '0' = not received vs a notable share receiving reminders), yet the no-show rate persists, reinforcing that reminder timing/frequency — not just reminder presence — needs to be redesigned.
- The neighbourhood breakdown confirms Jardim Camburi alone drives roughly 7K+ appointments, meaning targeted local interventions (e.g. community health worker follow-ups) in the top 3-4 neighbourhoods could meaningfully reduce overall no-show volume.
- Scholarship-linked appointments are a small fraction of total volume, so while their no-show rate is slightly elevated, the absolute impact on overall clinic capacity is modest compared to scheduling-driven factors like waiting time.

Health Condition Analysis Dashboard — Insights

- Patients with handicap level 4 show the highest no-show proportion (33.33%) among all handicap levels, despite being the smallest patient group — suggesting accessibility or transport barriers may be at play and could benefit from home-visit or telehealth alternatives.
- No-show proportions rise steadily with handicap severity (20.24% → 17.92% → 20.22% → 23.08% → 33.33% across levels 0-4), a gradient not obvious from the raw EDA bar chart alone.
- Female patients contribute roughly double the hypertension case volume of male patients, consistent with their larger overall share of appointments rather than a higher underlying hypertension rate.

- Chronic conditions (hypertension, diabetes, alcoholism) individually show only modest skew toward attendance or no-show, confirming that clinical risk factors are secondary to scheduling behavior as predictors of attendance.
- The age vs. appointment-count view shows the bulk of clinical volume sits with attended (not no-show) patients across the full age spectrum, meaning capacity-planning models should weight WaitingDays and neighbourhood more heavily than health status when forecasting no-show risk.

Taken together, the three dashboards reinforce the modeling result that scheduling and operational factors (waiting time, neighbourhood, month) outweigh demographic and clinical factors in predicting no-shows, and they pinpoint specific, actionable segments — long-wait bookings, high handicap-severity patients, and the top-volume neighbourhoods — where interventions are likely to have the greatest impact.

6. Key Findings

- Waiting time is the dominant driver of no-shows: the longer a patient waits between booking and the appointment date, the more likely they are to miss it. Same-day appointments have near-zero no-show rates.
- About 20% of all scheduled appointments in the dataset result in a no-show, representing a significant loss of clinical capacity.
- Younger adult patients show higher no-show tendencies compared to children and older patients.
- SMS reminders alone do not reduce no-shows in this dataset; this is likely confounded by reminders being sent more often for appointments booked far in advance, which independently raises no-show risk.
- Health conditions (hypertension, diabetes, alcoholism, handicap) have only a minor influence on attendance behavior compared to scheduling-related factors.
- The Decision Tree model, while accurate overall (79.8%), struggles to identify actual no-show cases due to class imbalance, reflected in its very low recall (0.13%) for the minority class despite a moderate AUC of 0.72.
- Jardim Camburi, Maria Ortiz, and Resistência neighbourhoods account for the highest volume of appointments and therefore the largest absolute number of no-shows.

7. Recommendations

- Apply class-balancing techniques (e.g. SMOTE, class-weighting, or undersampling) and try ensemble models (Random Forest, Gradient Boosting, XGBoost) to substantially improve recall on the no-show class.
- Introduce dynamic overbooking or stand-by scheduling for appointments with long waiting times, since these carry the highest no-show risk.
- Redesign the reminder strategy: send multiple, well-timed reminders (e.g. 48 hours and 2 hours prior) rather than a single SMS, especially for appointments booked far in advance.
- Target younger adult patients with appointment-confirmation calls or incentives, as this segment shows elevated no-show behavior.
- Use the WaitingDays feature as a real-time risk score at the point of booking, allowing staff to flag high-risk appointments for proactive follow-up.
- Continue monitoring no-show trends via the Power BI dashboards on a monthly basis to evaluate the impact of any intervention.

8. Conclusion

This project demonstrates an end-to-end healthcare analytics workflow: from data cleaning and exploratory analysis, through interactive Power BI dashboard design, to predictive modeling with a Decision Tree classifier. The analysis clearly identifies appointment waiting time as the single most influential factor behind patient no-shows, far outweighing demographic or health-related variables.

While the current model achieves good overall accuracy, its low recall for the no-show class highlights an important next step: addressing class imbalance to build a model that can proactively flag at-risk appointments, ultimately helping healthcare providers reduce missed appointments and use clinical resources more efficiently.